B.TECH · LABORATORY DEMONSTRATION

Fusion Splicing, OTDR Testing & Link Loss Measurement

Optical Fiber Communication Laboratory

A Fusion splicing **B** OTDR testing **C** Link loss

Student name	*Your student name here*
Roll number	*Your roll number here*
Department	*Your department here*
College	*Your college here*
Faculty	*Your faculty here*

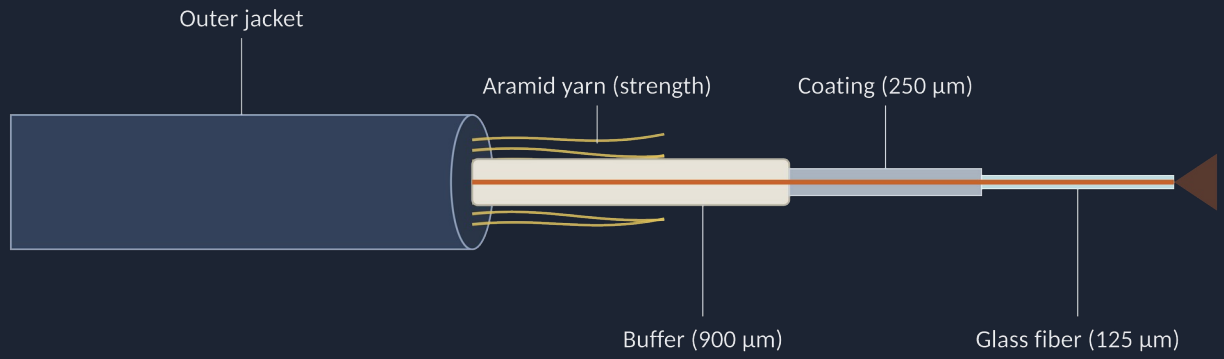
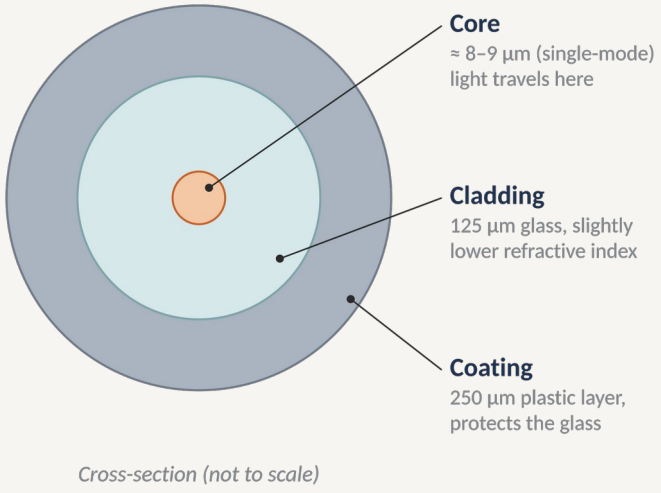
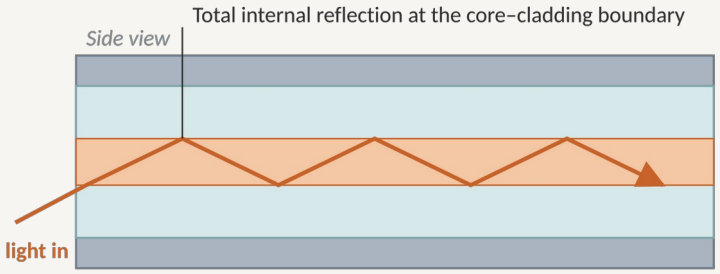


Illustration — layers not to scale

Optical Fiber and Fusion Splicing



Cross-section (not to scale)



Core index n_1 is a little higher than cladding index n_2. Light hitting the boundary at a shallow angle bounces back into the core, so it stays trapped and travels long distances.

Light travels in the tiny core. The cladding has a slightly lower refractive index, so light keeps reflecting back into the core (total internal reflection).

What is splicing?

Joining two fibers end-to-end so light passes from one core into the other.

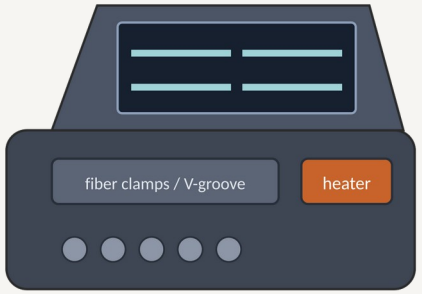
Fusion splicing

An electric arc melts the two glass ends into one piece. The joint is permanent, with very low loss (typically below 0.1 dB).

Why alignment matters

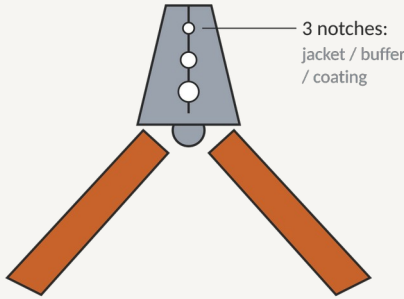
The core is only about 9 µm wide. Dirty, angled or offset ends lose light at the joint.

Equipment Required for Fusion Splicing



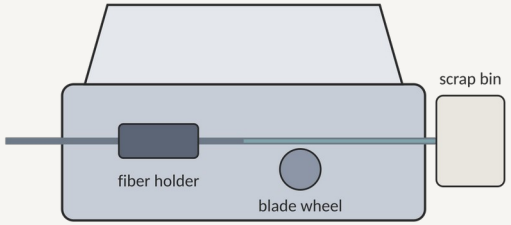
Fusion splicer

Aligns the two fibers with cameras and motors, then fuses them with an electric arc.



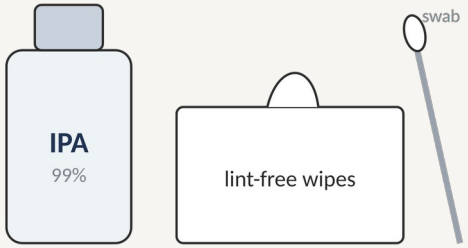
Fiber stripper

Removes the jacket, buffer and 250 µm coating without scratching the glass.



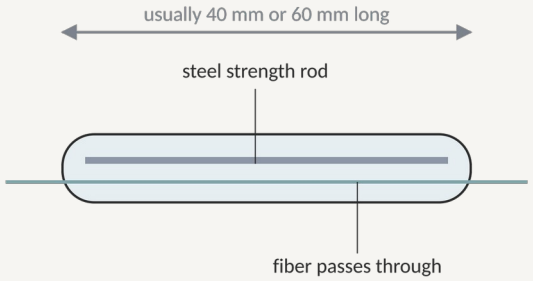
Precision cleaver

Cuts the bare fiber so the end is flat and at 90° to its length.



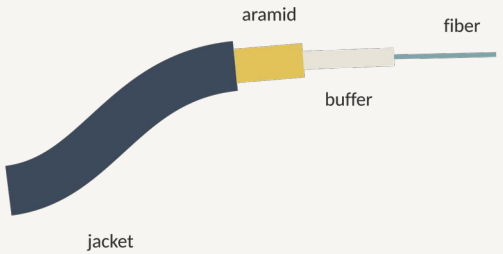
Cleaning kit

99% isopropyl alcohol and lint-free wipes remove dust and oil from the glass.



Heat-shrink sleeve

A tube with a steel rod that shrinks over the joint to protect it.



Optical fiber cable

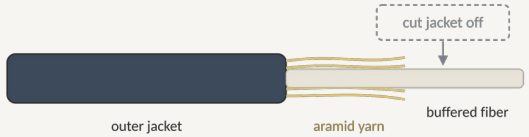
The fiber to be joined; check its type before you start.

Fusion Splicing Procedure (1/2)



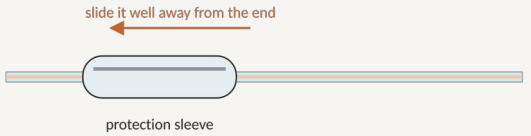
1 Prepare the cable

Check the fiber type and mark how much jacket to remove.



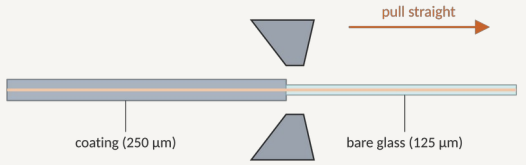
2 Remove the jacket

Strip the outer jacket; cut the aramid yarn with scissors.



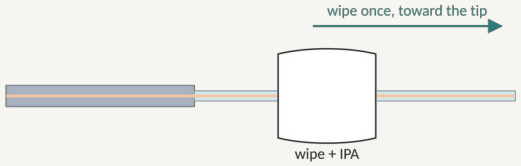
3 Slide on the sleeve

Put the heat-shrink sleeve on one fiber now — it can't be added later.



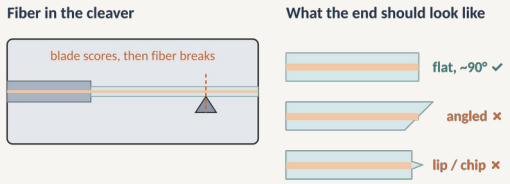
4 Strip the coating

Remove the 250 µm coating in one smooth pull (about 3–4 cm).



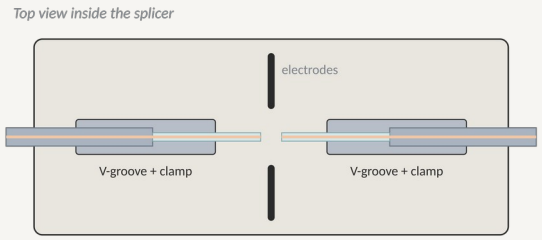
5 Clean the fiber

Wipe the bare glass once with 99% IPA on a lint-free wipe.



6 Cleave the ends

Cut a flat, square end (within about 1°). Bin the offcut.

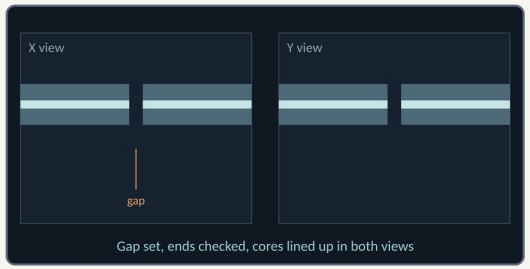


7 Load the splicer

Lay both fibers in the V-grooves, tips near the electrodes. Close the clamps.

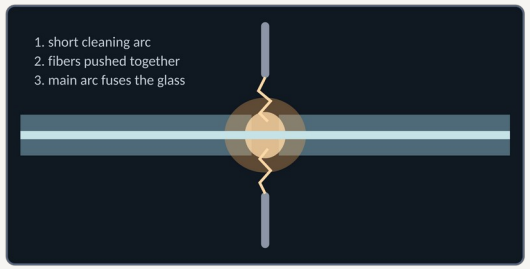
Safety: bare-fiber offcuts are tiny glass needles. Pick them up with tape and put them in the scrap bin.

Fusion Splicing Procedure (2/2)



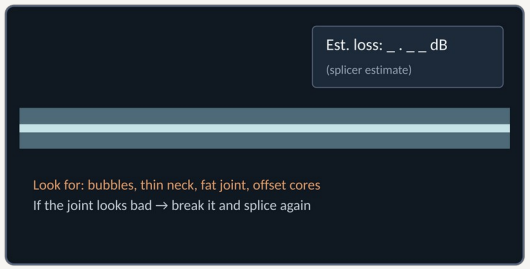
8 Align the cores

Cameras view the fibers in X and Y; the splicer lines up the cores.



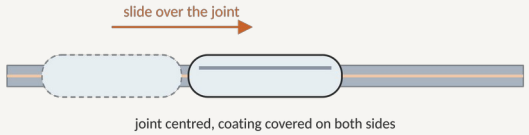
9 Fuse with the arc

A short cleaning arc, then the main arc melts the ends into one.



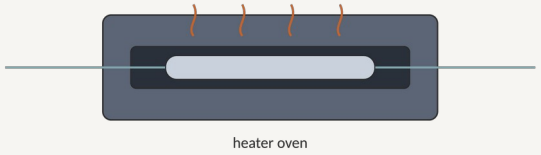
10 Inspect the splice

Check the joint on screen. The loss shown is only an estimate.



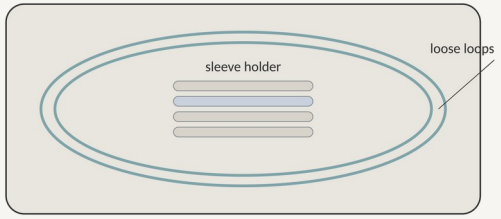
11 Move the sleeve

Slide the sleeve so the joint sits in its centre.



12 Heat-shrink

Shrink the sleeve in the heater. Let it cool before handling.



13 Splice tray

Fix the sleeve in the tray; coil spare fiber in large loops.

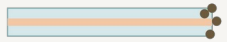
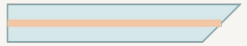
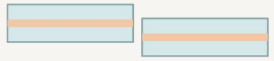
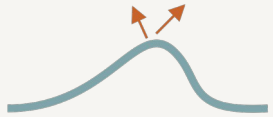
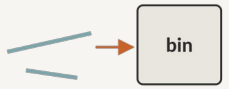


14 Test the joint

Measure the real loss with an OTDR or a power meter.

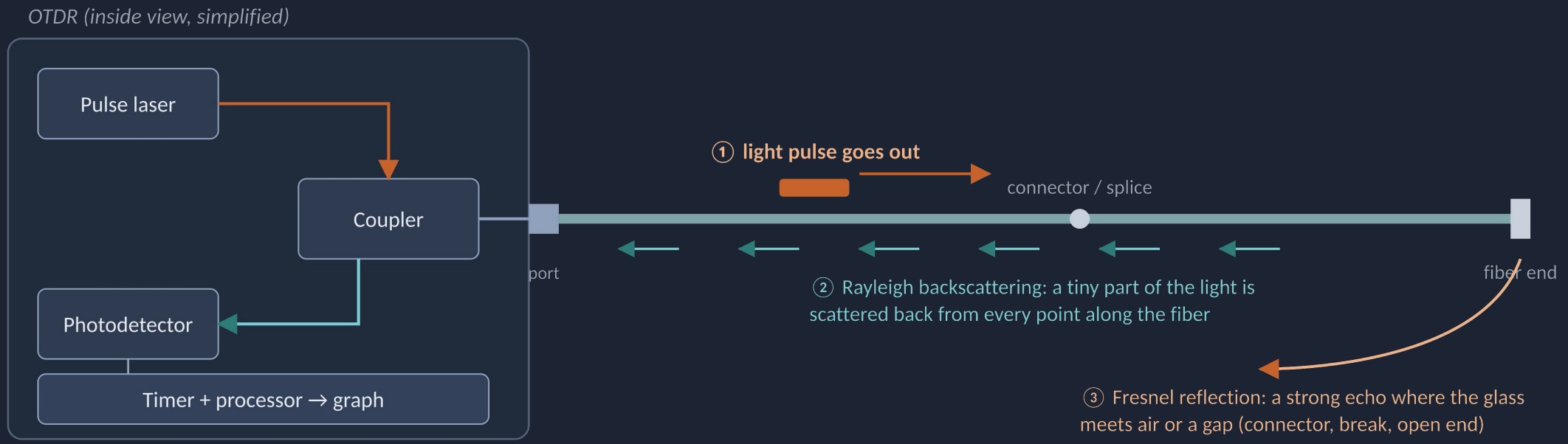
The splicer's loss value is an estimate from its camera image. Record the measured loss from step 14. (Screen views are illustrations; we did not record a splice-loss value.)

Splicing Precautions and Common Mistakes

	PROBLEM		WHAT TO DO
 dust on the end face	**Dirty fiber ends** Dust or skin oil burns in the arc and causes bubbles and high loss.	→	Clean with 99% IPA just before cleaving. Do not touch bare glass.
 angled end face	**Poor cleaving** Angled, chipped or lipped ends do not fuse well.	→	Keep the blade clean, rotate it when worn, and re-cleave any bad end.
 cores offset	**Incorrect alignment** If cores are offset, part of the light misses the next core.	→	Seat fibers properly in the V-grooves; clean the grooves; let the splicer align.
 light leaks at a tight bend	**Sharp fiber bends** Light leaks out at tight bends (worse at 1550 nm).	→	Keep large loops in the tray and follow the bend-radius guides.
 glass slivers → sealed bin	**Unsafe fiber fragments** Glass slivers can enter skin and eyes.	→	Wear safety glasses, use a scrap bin, don't eat or drink at the bench.

What Is an OTDR and How It Works

OTDR = Optical Time Domain Reflectometer. It finds the **length, losses and faults** of a fiber from just one end — it sends light pulses and times the light that comes back.

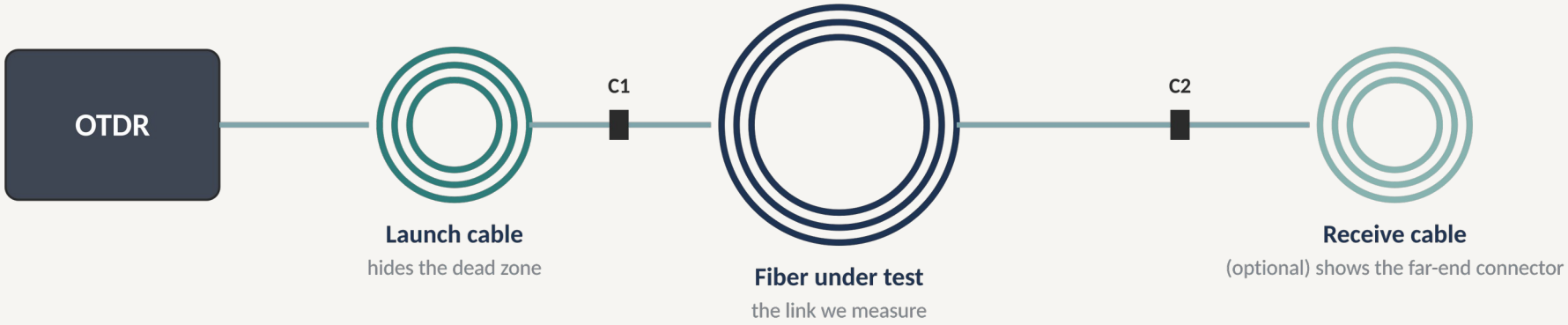


④ **Time of the echo → distance:** $d = (c \times t) / (2\,n)$

c = speed of light in vacuum, t = round-trip time, n = group index of the fiber (≈ 1.47). Divide by 2 because light goes out and back.

Connecting and Running the OTDR Test

Connection order



Connect

1. Switch on the OTDR.
2. Inspect and clean all connectors.
3. Connect launch cable (if available), then the fiber under test.
4. Check connections are firm.
5. Make sure no live signal is on the fiber.

Run the test

1. Open the Test Setting menu.
2. Set wavelength, range, pulse width, time, resolution, power mode.
3. Return to the main screen.
4. Press AVG to start the averaged test.
5. Wait, then read the trace and event table.

Why launch / receive cables?

The strong reflection at the OTDR port hides the first few metres (dead zone). A launch cable moves the first connector out of it; a receive cable lets the last connector be measured.

⚠️ **Never look into the OTDR port or a fiber end — the laser light is invisible.**

OTDR Test Settings We Used

Parameter	Our value	What it means
Test wavelength selection	**1310/1550 nm**	Wavelength(s) of laser light used for the test
Test range	**10 km**	Longest distance shown; set longer than the fiber
Pulse width	**50 ns**	Pulse length: short = fine detail, long = reaches farther
Test mode	**Manual**	We chose the settings ourselves (not Auto)
Test time	**50 seconds**	Averaging time: longer gives a smoother trace
Resolution	**Standard**	How finely the trace points are spaced
Launch cable setting	**None**	No launch cable length entered or removed
End cable setting	**None**	No receive (end) cable entered
Power mode	**Standard**	Instrument power option, left at default

Short pulse + long averaging = a clear, detailed trace for a short (few-km) fiber like ours.

LAB PHOTO

Photo 1 — test-settings screen

Insert the original photograph here (right-click → Change Picture), or place it in photos/ and rebuild.

Values copied exactly from our settings screen (Photo 1).

Understanding the OTDR Graph

LAB PHOTO

Photo 2 — our OTDR trace (1310 nm)

Insert the original photograph here (right-click → Change Picture), or place it in photos/ and rebuild.

- **X-axis**
  Distance along the fiber, in kilometres.
- **Y-axis**
  Level of the returned signal, in dB.
- **Sloping straight line**
  Normal fiber attenuation — the light weakens a little every km.
- **Sudden downward step**
  A loss event: possibly a splice, bend or other loss point.
- **Upward spike**
  A reflection: usually a connector or the fiber end.
- **Sharp drop to noise**
  End of the measured fiber link.

We only mark events that the OTDR actually shows on our screen and event table. We do not guess faults that are not on the trace.

Our OTDR Observations

Parameter	Observed value
Test wavelength	**1310 nm**
Pulse width	**50 ns**
Fiber distance	**4.822 km**
Total loss	**1.550 dB**
Displayed attenuation slope	**0.317 dB/km**
Optical Return Loss (ORL)	**29.37 dB**
Instrument result	**PASS**

All values read directly from the result screen (Photo 2). Nothing here is calculated.

What PASS means: the measured results were within the thresholds set in the instrument. It is not proof that the link meets every possible performance requirement.

LAB PHOTO

Photo 2 — result screen

Insert the original photograph here (right-click → Change Picture), or place it in photos/ and rebuild.

Link Loss: Meaning and Formulas

Link loss is the total drop in optical power as light travels through a fiber link (fiber + splices + connectors). Fiber attenuation is the loss per km.

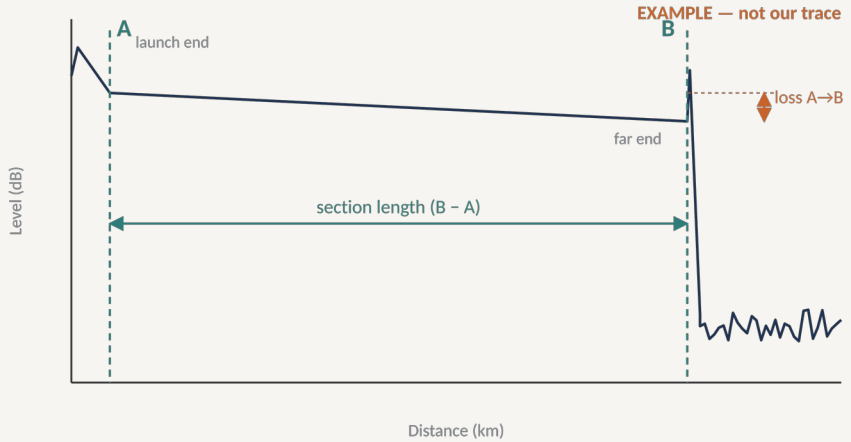
$$\text{Link Loss (dB)} = P_{in}\ \text{(dBm)} - P_{out}\ \text{(dBm)}$$

End-to-end insertion loss — measured with a light source and an optical power meter.

$$\text{Attenuation (dB/km)} = \textbf{Total Loss (dB) / Length (km)}$$

Average loss for every kilometre of the link.

An OTDR does not use the first formula. It estimates loss from backscattered light and gives the section loss (dB) in its event table or between the A–B cursors.



With the OTDR

1. Run the averaged (AVG) test.
2. Put cursor A at the start and B at the end of the link.
3. Read the A–B loss in dB and the length in km.
4. Divide loss by length for dB/km.

Trace above is an example, not our measurement.

Calculation Using Our Readings

GIVEN (measured)

Fiber length, L = **4.822 km**

Total loss = **1.550 dB**

CALCULATION

Average attenuation = Total loss / L

= 1.550 dB / 4.822 km

≈ 0.3214 dB/km

0.3214

dB/km · calculated by us
(total loss ÷ length)

0.317

dB/km · attenuation slope
shown by the OTDR (measured)

Why are they slightly different?

- The OTDR's slope is a straight-line fit over a smooth part of the trace. It leaves out events such as splices and connectors.
- Total loss covers the whole section, so any event loss inside it is included.
- So 0.317 is NOT simply 1.550 ÷ 4.822 — the two numbers come from different methods.

Difference = 0.3214 − 0.317 ≈ 0.004 dB/km (about 1.4%). Check: 0.317 dB/km × 4.822 km ≈ 1.529 dB, which is about 0.02 dB less than the measured 1.550 dB.

Results and Conclusion

4.822 km

fiber length · at 1310 nm

1.550 dB

total loss · at 1310 nm

PASS

instrument result · at 1310 nm

Discussion

- Average attenuation ≈ 0.321 dB/km (calculated); OTDR slope 0.317 dB/km.
- PASS only means the results were inside the instrument's set thresholds.
- Launch/end cables were 'None', so the first and last connector losses are not accurate.

Conclusion

1. Fusion splicing joins fibers permanently using an electric arc; clean, square cleaves give low loss.
2. An OTDR measures length and loss and locates faults from one end of the fiber.
3. Link loss is read from the OTDR trace; our link: 1.550 dB over 4.822 km at 1310 nm.

References: G. Keiser, Optical Fiber Communications (McGraw-Hill) · J. M. Senior, Optical Fiber Communications (Pearson) · FOA Online Reference Guide, thefoa.org/tech/ref · ITU-T G.652 · TIA-568.3-D · lab OTDR and splicer manuals.

Thank You

Questions?